CSE 219 — Signals and Linear Systems

Required Proofs · Anik Sir's Part (Lectures 1–7)

Detailed proofs · what to write · what is intuition only (Anik Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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How to Use This Sheet — status tags and exam writing style
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> **NOTE** Anik Sir: very long proofs that build a theory (FS → FT, how convolution and the Fourier-series equation are derived) are for intuition only; short proofs must be written; slides marked (*) and Parseval's theorem are not in the syllabus. Each proof below has a status tag. Slide references are PDF pages of the lecture files.

How to Use This Sheet

Tag	Meaning
REQUIRED	A short proof the slides do. Be able to write it in 3–8 lines.
REQUIRED (old papers)	Not a slide proof but asked in past TF papers (CSE 311 / EEE) on the same content. Learn it as well.
INTUITION	Long foundation derivation. Read for understanding; do not memorise for writing. A short version is given.

> **NOTE** Write proofs in the style shown: state what you prove, give the key substitution, end with the result. Rough sketches help (even–odd, convolution). Do not expand everything formally.

Lecture 1 — Signals

P1. Every signal has an even and an odd part — REQUIRED (sl. 43–44)

Claim. For any x(t), define $x_e(t) = ½[x(t) + x(−t)]$ and $x_o(t) = ½[x(t) − x(−t)]$. Then x_e is even, x_o is odd and $x = x_e + x_o$.

```
Even:  xₑ(−t) = ½[x(−t) + x(−(−t))] = ½[x(−t) + x(t)] = xₑ(t)
Odd:   xₒ(−t) = ½[x(−t) − x(t)] = −½[x(t) − x(−t)] = −xₒ(t)
Sum:   xₑ(t) + xₒ(t) = ½[x(t) + x(−t)] + ½[x(t) − x(−t)] = x(t)
```

Uniqueness (extra, 2 lines). If x = a + b with a even and b odd, then x(−t) = a(t) − b(t). Adding/subtracting x(t) and x(−t) gives $a = x_e$, $b = x_o$. The same proof works for x[n] with n in place of t.

Example (sl. 45–48). x = t + 1 on [−1, 1]: x(−t) = 1 − t, so $x_e = ½[(t+1) + (1−t)] = 1$ and $x_o = ½[(t+1) − (1−t)] = t$. Check by graph: x_e is a flat line (mirror-symmetric), x_o is a straight line through the origin (point-symmetric).

P2. The recipe for x(αt + β) — REQUIRED (sl. 26–31)

Claim. x(t) → x(t+β) → (reverse if α < 0) → scale by |α| gives x(αt+β), and the order cannot be swapped.

```
Step 1: g₁(t) = x(t + β)                        (shifts the picture left by β)
Step 2: g₂(t) = g₁(−t) = x(−t + β)   (only if α < 0)
Step 3: g₃(t) = g₂(|α|t).  If α < 0:  g₂(|α|t) = x(−|α|t + β) = x(αt + β).   If α > 0:  g₁(αt)
= x(αt + β).
Order matters: first compress then shift gives g(t) = x(2t) → g(t + 3) = x(2(t + 3)) = x(2t +
6) ≠ x(2t + 3).
Where does a feature go?  If x has a feature at t = a, then αt + β = a ⇒ t = (a − β)/α in x(αt
+ β).
```

Why: each operation replaces the variable t of the *current* signal. Shifting after scaling therefore shifts the argument of the scaled signal, not of the original.

Lecture 2 — Useful Signals

P3. Step and impulse are related — REQUIRED (sl. 11–15; CT: asked in CSE 311 2019-20 Q2(b))

Discrete time.

```
(a) δ[n] = u[n] − u[n−1]:  for n ≥ 1 both are 1 → 0;  n = 0: 1 − 0 = 1;  n < 0: 0 − 0 = 0.
This is exactly δ[n].
(b) u[n] = Σ_{m=−∞}^{n} δ[m]:  if n < 0 the sum window (−∞, n] does not contain m = 0 ⇒ 0;
    if n ≥ 0 it contains the single spike at m = 0 ⇒ 1.  So the sum equals u[n].
(c) u[n] = Σ_{k=0}^{∞} δ[n − k]:  δ[n−k] is 1 only at n = k; at a given n ≥ 0 exactly one term
(k = n) is 1; for n < 0 none.
```

Continuous time (short form). The step jumps from 0 to 1 at t = 0, so its derivative is 0 everywhere except at the jump. Replace the jump by a ramp of width Δ: the derivative is a pulse of height 1/Δ and width Δ, area 1 for every Δ. As Δ → 0 this becomes δ(t), so δ(t) = du/dt. Conversely $u(t) = \int_{-\infty}^{t}\delta(\tau)d\tau$: for t < 0 the window misses the impulse (0), for t > 0 it contains it (area 1). (The slides' detailed smoothed-step derivation, sl. 20–25, is marked (*) — excluded; the 4-line version above is enough.)

P4. Sifting property of the impulse — REQUIRED (Lec 6, sl. 45–51; asked in CSE 311 2012-13 Q5(a))

```
Since δ(t − a) = 0 for t ≠ a, only the value of x at t = a matters:  x(t)δ(t − a) = x(a)δ(t −
a).
Integrate:  ∫ x(t)δ(t − a)dt = x(a)∫δ(t − a)dt = x(a)   − provided a lies inside the
integration limits;
if a is outside [L, R] the impulse is never seen and the integral is 0.
Discrete: Σ_n x[n]δ[n − a] = x[a] (every other term is 0).
```

Consequence used constantly: δ(t − a) * h(t) = h(t − a); a train of impulses puts one copy of h at every impulse.

P5. Discrete-time periodicity condition — REQUIRED (sl. 43–50)

```
x[n] = cos(ω₀n + ϕ) is periodic with period N iff x[n + N] = x[n] for all n.
cos(ω₀(n + N) + ϕ) = cos(ω₀n + ϕ)  for all n  ⇔  ω₀N = 2πm for some integer m.   (the phase ϕ
plays no role)
So N = 2πm/ω₀ must be an integer ⇒ 2π/ω₀ = N/m must be a rational number.
The fundamental period N₀ is the numerator of 2π/ω₀ written in lowest terms.
Examples:  ω₀ = 2:  2π/2 = π (irrational) → not periodic.   ω₀ = 16π/63:  2π/ω₀ = 63/8 → N₀ =
63 (m = 8).
           ω₀ = 5/7:  2π/ω₀ = 14π/5 (irrational) → not periodic.
```

Sums: $x_1[n] + x_2[n]$ is periodic with $N = \text{lcm}(N_1, N_2)$. Continuous time: $T_0 = 2\pi/|\omega_0|$ always exists for a single sinusoid, but a sum is periodic only if the ratio T_1/T_2 is rational.

Lecture 3 — LTI Systems and Convolution

P6. The output of an LTI system is y = x * h — INTUITION (short version REQUIRED) (sl. 47–57; asked in CSE 311 2018-19 Q4(b))

Anik Sir treats the general derivation as intuition, but a time-domain proof is a 4-line argument and an old paper asked it.

```
Let δ[n] → h[n]   (definition of the impulse response).
1. Decompose the input into shifted, scaled impulses:  x[n] = Σ_k x[k] δ[n − k].
2. Time invariance:  δ[n − k] → h[n − k].
3. Homogeneity:       x[k] δ[n − k] → x[k] h[n − k].
4. Additivity (the sum is a sum of inputs):   y[n] = Σ_k x[k] h[n − k] = (x * h)[n].
```

Continuous time: $x(t) = \int x(\tau)\delta(t-\tau)d\tau$ (sifting). Each piece $x(\tau)d\tau\cdot\delta(t-\tau)$ produces $x(\tau)d\tau\cdot h(t-\tau)$ by TI and homogeneity. Adding (integrating) gives $y(t) = \int x(\tau)h(t-\tau)d\tau$. Only linearity and time invariance were used — so the formula holds for every LTI system.

Example (sl. 47–51). $x = \delta[n] + 2\delta[n-1] - \delta[n-2] + \delta[n-3]$, $h = u[n]$: $y = u[n] + 2u[n-1] - u[n-2] + u[n-3]$, i.e. 1, 3, 2, 3, 3, 3, …

P7. Convolution is commutative (and distributes over addition) — REQUIRED (short) (sl. 29–30; Lec 4, sl. 41)

```
Commutativity:  (a * b)[n] = Σ_k a[k] b[n − k].  Put m = n − k (k = n − m; as k runs over all
integers so does m):
                = Σ_m a[n − m] b[m] = Σ_m b[m] a[n − m] = (b * a)[n].
Continuous time: y(t) = ∫x(τ)h(t − τ)dτ, put σ = t − τ:  = ∫x(t − σ)h(σ)dσ = (h * x)(t).
Distributivity:  x * (h₁ + h₂) = Σ_k x[k](h₁[n−k] + h₂[n−k]) = x * h₁ + x * h₂  (linearity of
the sum).
Polynomial view: A(z)B(z) = B(z)A(z), and the coefficient of zⁿ is Σ_k a[k]b[n−k] — this is why
convolution is polynomial multiplication.
```

Useful consequence: pick the easier signal to flip; split a two-sided input into left- and right-sided parts and convolve separately (Lec 4, Ex 2.10).

Lecture 4 — Convolution, Causality and Stability

P8. Causal ⇔ h(t) = 0 for t < 0 — REQUIRED (sl. 55–56)

```
y(t) = ∫ h(τ)x(t − τ)dτ.  The input at time t − τ with τ < 0 is x at a time later than t (the
future).
(⇐) If h(τ) = 0 for τ < 0 the integral runs over τ ≥ 0 only: y(t) uses x(t − τ) with t − τ ≤ t
— past and present only ⇒ causal.
(⇒) h is the response to δ(t); a causal system cannot respond before the impulse arrives at t
= 0 ⇒ h(t) = 0 for t < 0.
DT: causal ⇔ h[n] = 0 for n < 0.
```

P9. BIBO stability: stable ⇔ ∫|h(t)|dt < ∞ — REQUIRED (sl. 57–59)

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Sufficiency.  Let |x(t)| ≤ B for all t.  Then
   |y(t)| = |∫h(τ)x(t − τ)dτ| ≤ ∫|h(τ)||x(t − τ)|dτ ≤ B ∫|h(τ)|dτ = B·M < ∞,   M = ∫|h|.
(bounded input → bounded output)
Necessity (idea, 2 lines).  If ∫|h| = ∞, choose the bounded input x(−τ) = sgn h(τ) (|x| ≤ 1).
Then
   y(0) = ∫h(τ)x(−τ)dτ = ∫|h(τ)|dτ = ∞  ⇒ the output is unbounded for a bounded input ⇒ not
stable.
DT:  stable ⇔ Σ_n |h[n]| < ∞.
```

Slide examples: $h_1 = e^{2t}u(-2-t)$: $\int_{-\infty}^{-2} e^{2t}dt = e^{-4}/2 < \infty$ stable (non-causal, since nonzero for $t \le -2$). $h_2 = te^{-t}u(t)$: $\int_0^{\infty} te^{-t}dt = 1 < \infty$ stable and causal. $e^{-4t}u(3-t)$ is non-causal and $\int_{-\infty}^{3} e^{-4t}dt = \infty \Rightarrow$ unstable.

P10. A nonzero periodic h cannot be causal — REQUIRED (sl. 60–61)

```
Let h be periodic with period T and h(t₀) ≠ 0 for some t₀ (≥ 0 w.l.o.g.).
Periodicity gives h(t₀) = h(t₀ − T) = h(t₀ − 2T) = …  Choose k with t₀ − kT < 0: then h is
nonzero at a negative time.
Causality needs h(t) = 0 for t < 0 — contradiction.  So only h ≡ 0 is both causal and periodic.
```

P11. A nonzero periodic h cannot be stable — REQUIRED (sl. 60, 62)

```
If h ≢ 0 is periodic, the area of |h| over one period is some A > 0 (same in every period).
∫_{−∞}^{∞}|h(t)|dt = … + A + A + A + … = ∞  ⇒ not absolutely integrable ⇒ not stable (by P9).
So only h ≡ 0 is both stable and periodic.
```

Lecture 5 — Fourier Series

P12. The Fourier-series coefficient formula $a_k = (1/T)\int x(t)e^{-jk\omega_0 t}dt$ — INTUITION (short version REQUIRED) (sl. 18–22)

Given $x(t) = \Sigma_m a_m e^{jm\omega_0 t}$, how do we get a_k? Stop the k-th wave rotating and average.

```
Key fact:  (1/T)∫_T e^{jnω₀t}dt = 1 if n = 0, and = (1/T)[e^{jnω₀t}/(jnω₀)]_{t₀}^{t₀+T} =
(1/T)(e^{j2πn} − 1)·(…) = 0 if n ≠ 0.
Multiply x(t) by e^{−jkω₀t}:   x(t)e^{−jkω₀t} = Σ_m a_m e^{j(m − k)ω₀t}.
Average over one period:        (1/T)∫_T x(t)e^{−jkω₀t}dt = Σ_m a_m·(1/T)∫_T e^{j(m − k)ω₀t}dt
= a_k   (only m = k survives).
Special case k = 0:  a₀ = (1/T)∫_T x(t)dt = the average value of x.
```

Memorise the *result* and the *averaging idea*; the full justification (swapping sum and integral, convergence) is not required.

P13. Coefficients by inspection and the DC term — REQUIRED (sl. 25–29, 36–37)

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cos θ = (e^{jθ} + e^{−jθ})/2,  sin θ = (e^{jθ} − e^{−jθ})/(2j).  Write every term with
e^{jkω₀t}, collect equal k.
Example: x = 1 + sin ω₀t + 2cos ω₀t + cos(2ω₀t + π/4)
  sin ω₀t + 2cos ω₀t = (1/2j)(e^{jω₀t} − e^{−jω₀t}) + (e^{jω₀t} + e^{−jω₀t}) = (1 −
j/2)e^{jω₀t} + (1 + j/2)e^{−jω₀t}
  cos(2ω₀t + π/4) = ½e^{jπ/4}e^{j2ω₀t} + ½e^{−jπ/4}e^{−j2ω₀t}
  a₀ = 1, a₁ = 1 − j/2, a₋₁ = 1 + j/2, a₂ = ½e^{jπ/4} = (√2/4)(1 + j), a₋₂ = (√2/4)(1 − j).
Geometry for a₀ (sl. 37): a₀ = (area over one period)/T, e.g. triangular ramp T = 3: (½·3·1)/3
= ½.
```

P14. Fourier series of a rectangular pulse train — REQUIRED (sl. 30–34)

```
x = 1 for |t| < T₁, 0 for T₁ < |t| < T/2 (period T, ω₀ = 2π/T).  Use the period (−T/2, T/2):
k = 0:  a₀ = (1/T)∫_{−T₁}^{T₁}1 dt = 2T₁/T
k ≠ 0: a_k = (1/T)∫_{−T₁}^{T₁}e^{−jkω₀t}dt = (1/T)[e^{−jkω₀t}/(−jkω₀)]_{−T₁}^{T₁} =
(1/T)(e^{jkω₀T₁} − e^{−jkω₀T₁})/(jkω₀)
      = 2 sin(kω₀T₁)/(Tkω₀) = sin(kω₀T₁)/(kπ)      (since Tω₀ = 2π)
T₁ = T/4:  a₀ = ½, a_k = sin(kπ/2)/(kπ): a±₁ = 1/π, a±₂ = 0, a±₃ = −1/(3π), a±₅ = 1/(5π), …
```

P15. Even signals have real coefficients, odd signals imaginary; half a cycle is enough — REQUIRED (old papers: CSE 311 2018-19 Q2(b), 2022-23 Q1(a))

```
a_k = (1/T)∫_{−T/2}^{T/2} x(t)[cos kω₀t − j sin kω₀t]dt
Even x:  x·sin kω₀t is odd ⇒ its integral over the symmetric period is 0, and x·cos kω₀t is
even:
         a_k = (2/T)∫₀^{T/2} x(t)cos kω₀t dt   — real (and a₋ₖ = a_k).
Odd x:   x·cos kω₀t is odd ⇒ 0, and x·sin kω₀t is even:
         a_k = −(2j/T)∫₀^{T/2} x(t)sin kω₀t dt  — purely imaginary (a₋ₖ = −a_k), and a₀ = 0.
cos 2πf₀t (even) → a±₁ = ½ real;  sin 2πf₀t (odd) → a±₁ = ∓j/2 imaginary.
```

Lecture 6 — Properties of Fourier Series

All of these follow by substituting into $x(t) = \Sigma\, a_k\, e^{jk\omega_0 t}$. Each is a 2–3 line proof — all REQUIRED.

P16. Linearity, time shift, time reversal, time scaling (sl. 5–35)

```
Linearity:  αx + βy = Σ(αa_k + βb_k)e^{jkω₀t}   (same period, same exponentials, so
coefficients add).
Time shift: x(t − t₀) = Σ a_k e^{jkω₀(t − t₀)} = Σ [a_k e^{−jkω₀t₀}] e^{jkω₀t}   ⇒ b_k = a_k
e^{−jkω₀t₀}.
            |e^{−jkω₀t₀}| = 1: the magnitude is unchanged; the phase turns by −kω₀t₀ (higher
harmonics rotate more).
Reversal:   x(−t) = Σ a_k e^{jkω₀(−t)} = Σ a_k e^{j(−k)ω₀t} = Σ a_{−m} e^{jmω₀t} (m = −k)   ⇒ b_k
= a_{−k}.
Scaling:    x(αt) = Σ a_k e^{jk(αω₀)t}: same coefficients a_k, but the fundamental frequency
becomes αω₀ (period T/α).
Combined (sl. 24–35): x(4 − t) + y(t − 3) ↔ a_{−k}e^{−j4kω₀} + b_k e^{−j3kω₀};   x(4 − 3t) ↔
a_{−k}e^{−j4kω₀}, ω₀′ = 3ω₀
            (shift left 4 → a_k e^{j4kω₀};  reverse → a_{−k}e^{−j4kω₀};  scale → coefficients
unchanged).
```

TRAP Slide 26 prints $y(t-3) \leftrightarrow a_k\, e^{-j3k\omega_0}$; it must be **$b_k$** $e^{-j3k\omega_0}$ because y has coefficients b_k.

P17. Differentiation — REQUIRED (sl. 36–41, 57–63)

```
x(t) = Σ a_k e^{jkω₀t}  ⇒  dx/dt = Σ a_k d/dt[e^{jkω₀t}] = Σ (jkω₀ a_k) e^{jkω₀t}   ⇒  dx/dt ↔
jkω₀ a_k.
Use the current fundamental: for x(4 − 3t), ω₀′ = 3ω₀ ⇒ d/dt x(4 − 3t) ↔ j(3kω₀)a_{−k}e^{−j4kω₀}.
At a jump of height J at t₀ the derivative contains Jδ(t − t₀).
Check on the rectangular pulse train: dx/dt = δ(t + T₁) − δ(t − T₁) per period ⇒ b_k =
(1/T)(e^{jkω₀T₁} − e^{−jkω₀T₁}) = (2j/T)sin kω₀T₁
   and jkω₀a_k = jkω₀·sin(kω₀T₁)/(kπ) = j(ω₀/π)sin kω₀T₁ = (2j/T)sin kω₀T₁ ✓.
```

P18. Integration — REQUIRED (sl. 64–72)

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If y = ∫x dt is periodic, then dy/dt = x ⇒ a_k = jkω₀ b_k ⇒ b_k = a_k/(jkω₀) for k ≠ 0.
The constant of integration is invisible to the derivative, so b₀ must be found separately
(average value of y).
Example: sawtooth x = t on [0,1), T = 1: dx/dt = 1 − Σδ(t − m): coefficients c₀ = 0 (1 − 1),
c_k = −1 (k ≠ 0)
   ⇒ a_k = −1/(j2πk) = j/(2πk), a₀ = ∫₀¹t dt = ½.
```

P19. Multiplication ↔ convolution of the coefficients — REQUIRED (sl. 73–81)

```
x = Σ_p a_p e^{jpω₀t},  y = Σ_q b_q e^{jqω₀t}.   x·y = Σ_p Σ_q a_p b_q e^{j(p + q)ω₀t}.
The k-th harmonic collects all pairs with p + q = k, i.e. q = k − p:   c_k = Σ_p a_p b_{k−p} =
(a * b)_k.
Example: x = 1 + cos ω₀t, a = {¼, 1, ½} (k = −1,0,1):  a * a = {¼, 1, 3/2, 1, ¼} (k = −2…2).
Check: (1 + cos)² = 3/2 + 2cos ω₀t + ½cos 2ω₀t ⇒ c₀ = 3/2, c±₁ = 1, c±₂ = ¼ ✓.
```

P20. The impulse train has a flat line spectrum — a train in frequency — REQUIRED (sl. 52–56)

> x(t) = Σ_m δ(t − mT), ω₀ = 2π/T. Take the period (−T/2, T/2): it contains exactly one impulse, at t = 0.
> a_k = (1/T)∫_{−T/2}^{T/2} δ(t)e^{−jkω₀t}dt = (1/T)e^{−jkω₀·0} = 1/T (sifting, P4) for every integer k.
> So the coefficient sequence is the constant 1/T: lines of equal height 1/T at every kω₀ — an impulse train in frequency (spacing ω₀ = 2π/T).
> Trigonometric form: x = 1/T + (2/T)Σ_{n≥1}cos(2πnt/T).

Lecture 7 — Fourier Transform

P21. From Fourier series to Fourier transform — INTUITION ONLY (sl. 4–18)

Anik Sir: do **not** memorise this derivation; understand it. The story in six lines:

```
1. Repeat an aperiodic pulse x(t) with period T:  x̃_T(t) = Σ_m x(t − mT) — now periodic, so it
has a Fourier series.
2. a_k = (1/T)∫_{-T/2}^{T/2} x̃_T e^{-jkω₀t}dt = (1/T)X(jkω₀),   X(jω) = ∫x(t)e^{-jωt}dt  (the
pulse is zero outside one period).
3. So T·a_k are samples of one smooth curve X(jω) at ω = kω₀.
4. x̃_T(t) = Σ a_k e^{jkω₀t} = (1/2π)Σ X(jkω₀)e^{jkω₀t}Δω with Δω = ω₀ = 2π/T.
5. T → ∞: x̃_T → x(t); Δω → dω; kω₀ → ω; the sum → ∫.
6. x(t) = (1/2π)∫X(jω)e^{jωt}dω   and   X(jω) = ∫x(t)e^{-jωt}dt   — the transform pair.
```

P22. Frequency shift and cosine modulation — REQUIRED (sl. 22–28)

```
Frequency shift:  F{x(t)e^{jω₀t}} = ∫x(t)e^{jω₀t}e^{-jωt}dt = ∫x(t)e^{-j(ω − ω₀)t}dt = X(j(ω −
ω₀)).
                  (every component e^{jωt} becomes e^{j(ω + ω₀)t}: the whole spectrum slides
right by ω₀.)
Modulation:       x(t)cos ω₀t = ½x(t)e^{jω₀t} + ½x(t)e^{-jω₀t}  ⇒  ½X(j(ω − ω₀)) + ½X(j(ω +
ω₀)).
                  Two half-height copies of the spectrum, centred at +ω₀ and −ω₀.
```

P23. Transforms of the rectangle, the impulse, and the shifted versions — REQUIRED (sl. 29–44)

```
Rectangle (1 for |t| < T/2):  X(jω) = ∫_{-T/2}^{T/2}e^{-jωt}dt = [e^{-jωt}/(-jω)]_{-T/2}^{T/2}
= (e^{jωT/2} − e^{-jωT/2})/(jω)
                              = 2 sin(ωT/2)/ω         (X(0) = T).   Wider T → narrower main lobe
(first zero 2π/T).
Time shift:  F{x(t − t₀)} = ∫x(t − t₀)e^{-jωt}dt;  put τ = t − t₀ ⇒ e^{-jωt₀}∫x(τ)e^{-jωτ}dτ =
e^{-jωt₀}X(jω).
             |X| unchanged, phase −ωt₀ (a straight line in ω). Example: 1 on (4, 10) = width-6
rect shifted by 7 ⇒ e^{-j7ω}·2 sin 3ω/ω, X(0) = 6.
Impulse:     F{δ(t)} = ∫δ(t)e^{-jωt}dt = e^{-jω·0} = 1   (sifting)  — flat spectrum: a spike
contains all frequencies equally.
             F{δ(t − t₀)} = e^{-jωt₀}.
Lego block (1 on (1,4) plus 0.5 on (2,3), both centred at 2.5):  X = e^{-j2.5ω}[2 sin(1.5ω)/ω +
0.5·2 sin(0.5ω)/ω].
```

P24. Extra results asked in the papers on the same content — REQUIRED (old papers); not in the slides

These are not in Lecture 7, but 2022-23, 2023-24 and the older papers use them. Learn the short proofs.

Time delay ⇒ linear phase (CSE 311 2019-20 Q2(a)): $x(t - t_0) \leftrightarrow e^{-j\omega t_0}X(j\omega)$; phase $-\omega t_0$ is linear in ω.
If a voice (many frequencies) lost linear phase, each frequency would be delayed differently (dispersion) and the sound would smear.
Time scaling: $F\{x(at)\} = \int x(at)e^{-j\omega t}dt$; $\tau = at \Rightarrow (1/|a|)\int x(\tau)e^{-j\omega\tau/a}d\tau = (1/|a|)X(j\omega/a)$.
($|a|$ because $a < 0$ flips the integration limits.) Compression in time ⇒ expansion in frequency.
Differentiation: $x = (1/2\pi)\int X\, e^{j\omega t}d\omega \Rightarrow dx/dt = (1/2\pi)\int j\omega X\, e^{j\omega t}d\omega \Rightarrow dx/dt \leftrightarrow j\omega X(j\omega)$.
Convolution (CSE 311 2012-13 Q5(b), 2011-12 Q6(a)):
$F\{x*h\} = \int[\int x(\tau)h(t - \tau)d\tau]e^{-j\omega t}dt = \int x(\tau)[\int h(t - \tau)e^{-j\omega t}dt]d\tau$; $s = t - \tau \Rightarrow \int x(\tau)e^{-j\omega\tau}H(j\omega)d\tau = X(j\omega)H(j\omega)$.
$e^{-at}u(t)$, $a > 0$: $\int_0^\infty e^{-(a + j\omega)t}dt = 1/(a + j\omega)$. $e^{-a|t|} = e^{-at}u(t) + e^{at}u(-t) \leftrightarrow 1/(a + j\omega) + 1/(a - j\omega) = 2a/(a^2 + \omega^2)$.
Reversal: $x(-t) \leftrightarrow X(-j\omega)$ (put $\tau = -t$). $\mathrm{sgn}(t) = \lim_{a\to 0}[e^{-at}u(t) - e^{at}u(-t)] \leftrightarrow 1/(a + j\omega) - 1/(a - j\omega) \to 2/(j\omega)$.
Real x: $X(-j\omega) = \int x\, e^{j\omega t}dt = X^*(j\omega) \Rightarrow |X|$ even, $\angle X$ odd (CSE 311 2016-17 Q1(b)).
Duality: $2\pi x(-t) = \int X(j\omega)e^{-j\omega t}d\omega$; rename $t \leftrightarrow \omega \Rightarrow F\{X(jt)\} = 2\pi x(-\omega)$. (rect ↔ sinc ⇒ sinc ↔ $2\pi\cdot$rect.)

TRAP Not required: Parseval's theorem and anything marked (*). Do not spend time on them.